Robert Marlow

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EDUCATION

University of California, Los Angeles
Master of Science, Mechanical Engineering, expected June 2027
- GPA: 3.96 Coursework: Bionic System Design, Linear Optimal Control, Flexures, Soft Structures/Robots

University of Wisconsin-Madison
Bachelor of Science, Mechanical Engineering, May 2025
- GPA: 4.0 Coursework: Mechatronics, Component Design, Kinematics/Dynamics of Robot Manipulators

EXPERIENCE

Dropzone Robotics **Pasadena, CA**
Mechanical Design Intern Jun 2026-Present
- Mapped and resolved hardware tradeoffs for 47 component options across 6 sensing and processing needs
- Supported phase I and II SBIR/STTR contracts and new proposals with feasibility, risk, and failure mode analyses
- Designed, implemented, and validated a custom heatsink and field-deployable enclosure for wearable edge perception and compute, surpassing mass reduction goals by 36% for heatsink and 25% for full assembly
- Owned design, documentation, and full system testing of hybrid soft/rigid mobile swarm robots for patient transport

Mohala Lab @ UCLA **Los Angeles, CA**
Graduate Research Assistant Jan 2025 - Present
- Conceptualized, prototyped, and characterized grasping behavior of growing underactuated soft pneumatic grippers
- Investigated control strategies for soft legged locomotion by training reinforcement learning policies
- Mentored 4 undergraduate students' research projects, advising on design considerations and literature searches

UW-Madison BADGER Lab **Madison, WI**
Undergraduate Research Assistant Jan 2024 - May 2025
- Wrote and maintained ROS/ROS2 nodes for teleoperation and state estimation of a passively stable mobile robot
- Researched and deployed docker images to mitigate versioning conflicts and manage dependencies

Triton Medical Robotics **Burlingame, CA**
Product Development Engineer Intern May 2024 - Aug 2024
- Developed Python scripts to compile and analyze robot telemetry data from rosbags, extracting driving statistics from physician labs to underpin endoscope and capital equipment cyclic durability test protocols
- Advanced manufacturing capabilities by designing and implementing assembly line fixtures, drafting and updating process instructions, and investigating rejected builds to raise yields, identify training gaps, and inform specs

Engineering Intern May 2023 - Aug 2023
- Designed test fixtures and performed 2000+ trials to measure climate-controlled behavior of composite components
- Originated new product demonstration tools for compact pneumatic control, achieving long-term adoption
- Introduced and validated test methods using statistical analysis of new and historical data for catheter subassemblies

PROJECTS

Mars Rover Science Payload, Capstone Design for Wisconsin Robotics
- Built and iterated a remotely operated soil auger for mobile subsurface sampling in the University Rover Challenge
- Established and met weight, power, and space constraints based on competition rules and client input
- Conducted literature reviews and prototype testing to inform auger geometry and motor performance requirements
- Coordinated across teams to plan modular electromechanical integration with other science subsystems and rover

Lighted Turn Signal Glove, UW MadMakers
- Developed a wearable device to help improve visibility and communication when bicycling at night
- Implemented simple gesture and orientation recognition using an IMU sensor and ESP32 microcontroller
- Architected and built a custom PCB to hold LEDs, battery charging, ESP32, IMU, and UART/SPI connections

SKILLS

Software: Solidworks, Onshape, Inventor, KiCad, Minitab, Excel, OpenSim, EES, LaTeX
Programming: Python, Matlab, Java, C++, R, Arduino, ROS, ROS2, Docker, Linux
Tools: Shop Machines, 3D Printing, Soldering, Oscilloscope, Welding, Metal Casting, Glassworking
Design: DFM, GD&T, FMEA, Electrical Schematics, PCB Design, Component Selection